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MATHEMATICS

# Second Order Monotone Difference Schemes with Approximation on Non-Uniform Grids for Two-Dimensional Quasilinear Parabolic Convection-Diffusion Equations

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**Abstract**—The present communication is devoted to the construction of monotone difference schemes of the second order of local approximation on non-uniform grids in space for 2D quasi-linear parabolic convection-diffusion equation. With the help of difference maximum principle, two-sided estimates of the difference solution are established and an important a priori estimate in a uniform norm $C$ is proved. It is interesting to note that the maximal and minimal values of the difference solution do not depend on the diffusion and convection coefficients.

**Keywords:** non-uniform grid, maximum principle, regularization principle, monotone difference scheme, convection-diffusion equation.

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## 1. INTRODUCTION

In the theory of difference schemes [1, 2], the maximum principle is applied to study the stability and convergence of difference solutions in the uniform norm. Finite-difference methods satisfying the grid maximum principle are usually called monotone (see [1, 2]). Monotone schemes play an important role in computational practice. They make it possible to obtain numerical solutions without oscillations even in the case of non-smooth solutions [3].

When constructing monotone difference schemes, it is desirable to preserve the second order of approximation in a spatial variable. Such schemes were constructed for parabolic and hyperbolic equations with the presence of lower derivatives. For example, a non-conservative scheme of the second order of approximation for linear parabolic equations of general form on uniform grids was given in [1, 2]. In the case of non-uniform grids for stationary and non-stationary problems with variable coefficients without lower derivatives, a scheme was obtained in [4] for which the maximum principle is satisfied without restrictions on the relations of the coefficients and steps of the spatial grid (unconditional monotonicity for stationary equations). In the work [5], economical and unconditionally monotone schemes of the second order of approximation were constructed on a non-uniform grid for non-stationary multi-dimensional convection-diffusion problems.

The present note is devoted to the construction of monotone difference schemes of the second order of local approximation on non-uniform grids in space for a two-dimensional quasi-linear parabolic convection-diffusion equation. The construction of such schemes is based on the use of a special differential

identity and the principle of regularization [1, 2, 5]. Two-sided estimates [6] of the difference solution are established and an important a priori estimate is proved in the uniform norm $C$.

## 2. AUXILIARY RESULTS

Let $\Omega_h$ be a finite set of nodes (grid) in a bounded domain of $n$-dimensional Euclidean space and $x \in \Omega_h$ be a point of $\Omega_h$. Consider the equation

$$A(x)y(x) = \sum_{\xi \in \mathcal{M}'(x)} B(x,\xi)y(\xi) + F(x), \quad x \in \Omega_h, \tag{1}$$

which is called the canonical form of the difference scheme under consideration [1]. Here, $\mathcal{M}'(x) = \mathcal{M}(x)\backslash x$ and $\mathcal{M}(x)$ is the stencil of the scheme. Since any difference scheme can be written in the form of (1), by its monotonicity, we mean that the coefficients of the equation satisfy the positivity conditions

$$A(x) > 0, \quad B(x,\xi) > 0, \quad \forall \xi \in \mathcal{M}'(x), \tag{2}$$

$$D(x) = A(x) - \sum_{\xi \in \mathcal{M}'(x)} B(x,\xi) > 0, \quad \forall \xi \in \mathcal{M}'(x). \tag{3}$$

To derive a two-sided estimate for the solution of the difference scheme, it is convenient to use the following result.

**Lemma 1.** (*see* [7]) *Suppose that the positivity conditions* (2), (3) *for the coefficients are fulfilled. Then the maximal and minimal values of the solution of the difference scheme* (1) *belong to the following value interval of the input data:*

$$\min_{x \in \Omega_h} \frac{F(x)}{D(x)} \le y(x) \le \max_{x \in \Omega_h} \frac{F(x)}{D(x)}, \quad x \in \Omega_h. \tag{4}$$

**Corollary 1.** *Let the conditions of the lemma be fulfilled. Then, for the solution of the difference problem* (1), *the following estimate in the grid norm C holds:*

$$\|y\|_C = \max_{x \in \Omega_h} |y(x)| \le \left\| \frac{F}{D} \right\|_C.$$

## 3. PROBLEM STATEMENT AND DIFFERENCE SCHEME

Let $\bar{Q}_T = \bar{\Omega} \times [0 \le t \le T]$ be a rectangle with the boundary denoted by $\Gamma$. The task is to find a function $u(x,t)$, $x = (x_1, x_2)$, satisfying the initial boundary value problem for 2D quasi-linear parabolic convection-diffusion equation

$$\frac{\partial u}{\partial t} = \sum_{\alpha=1}^{2} \frac{\partial}{\partial x_\alpha}\left(k_\alpha(u)\frac{\partial u}{\partial x_\alpha}\right) + \sum_{\alpha=1}^{2} v_\alpha(u)\frac{\partial u}{\partial x_\alpha} - q(x)u + f(x,t), \quad x \in \Omega, \quad t \in (0,T], \tag{5}$$

with Dirichlet boundary conditions

$$\begin{aligned} &u(0,x_2,t) = \mu_1(x_2,t), \quad u(l_1,x_2,t) = \mu_2(x_2,t), \quad 0 \le x_2 \le l_2, \quad 0 < t \le T, \\ &u(x_1,0,t) = \mu_3(x_1,t), \quad u(x_1,l_2,t) = \mu_4(x_1,t), \quad 0 \le x_1 \le l_1, \quad 0 < t \le T, \end{aligned} \tag{6}$$

and the initial condition

$$u(x,0) = u_0(x), \quad x \in \bar{\Omega}. \tag{7}$$

We suppose that $q(x) \ge 0$, $x \in \Omega$ and there exist two real numbers $k_{\min}$ and $k_{\max}$ so that the condition of parabolicity of Eq. (5) on the solution is satisfied [8]

$$0 < k_{\min} \le k_\alpha(u) \le k_{\max}, \quad \forall u \in \bar{D}_u, \quad k_{\min}, k_{\max} = \text{const}, \quad \alpha = 1,2,$$
$$\bar{D}_u = \{u(x,t)\colon m_1 \le u(x,t) \le m_2,\ (x,t) \in \bar{Q}_T\},$$

with $m_1$ and $m_2$ are constants, defined by the conditions

$$m_1 = \min\{0, \mu_{\min}(x,t), \min_{x \in \Omega} u_0(x)\} + T \min\{0, \inf_{(x,t) \in Q_T} f(x,t)\},$$

$$m_2 = \max\{0, \mu_{\max}(x,t), \max_{x\in\bar\Omega} u_0(x)\} + T\max\{0, \sup_{(x,t)\in Q_T} f(x,t)\},$$

$$\mu_{\min}(x,t) = \min_{(x,t)\in\bar Q_T}\{\mu_1(x_2,t), \mu_2(x_2,t), \mu_3(x_1,t), \mu_4(x_1,t)\},$$

$$\mu_{\max}(x,t) = \min_{(x,t)\in Q_T}\{\mu_1(x_2,t), \mu_2(x_2,t), \mu_3(x_1,t), \mu_4(x_1,t)\}.$$

Further assume that problem (5)–(7) has a unique solution and all the coefficients involved in Eq. (5) and the sought function have continuous bounded derivatives of the required order.

In $\bar\Omega$ we introduce an arbitrary non-uniform grid

$$\hat{\bar\omega}_h = \{x_{i_1 i_2} = (x_1^{i_1}, x_2^{i_2}), x_\alpha^{i_\alpha} = x_\alpha^{i_\alpha - 1} + h_\alpha^{i_\alpha}, i_\alpha = 1,2,\ldots,N_\alpha - 1, x_\alpha^0 = 0, x_\alpha^{N_\alpha} = l_\alpha, \alpha = 1,2\},$$

where $\sum_{i_\alpha=1}^{N_\alpha} h_\alpha^{i_\alpha} = l_\alpha$ for $\alpha = 1,2$. Let $\hat\omega_h$ denote the set of internal nodes of $\hat{\bar\omega}_h$ and $\gamma_h$ denote the set of its boundary nodes. In addition, we will consider a non-uniform grid in time variable

$$\hat{\bar\omega}_\tau = \{t_0 = 0, t_n = t_{n-1} + \tau_n, n = 1,2,\ldots,N_0\},$$

i.e.,

$$\hat{\bar\omega}_\tau = \hat\omega_\tau \cup \{T\}, \quad \sum_{n=1}^{N_0}\tau_n = T.$$

For simplicity, we use non-index designations for independent variables

$$x = x_{i_1 i_2}, \quad x_\alpha = x_\alpha^{i_\alpha}, \quad x_{\alpha\pm} = x_\alpha^{i_\alpha \pm 1}, \quad h_\alpha = h_\alpha^{i_\alpha}, \quad h_{\alpha+} = h_\alpha^{i_\alpha+1}, \quad t = t_n, \quad \hat t = t_{n+1},$$

and for grid functions

$$g = g_{i_1 i_2} = g(x_1^{i_1}, x_2^{i_2}, t_n) = g(x,t), \quad g^{\pm 1_1} = g_{i_1\pm 1, i_2},$$
$$g^{\pm 1_2} = g_{i_1, i_2\pm 1}, \quad \hat g = g^{n+1} = g(x, t_{n+1}).$$

When constructing monotone schemes of the second order of accuracy on a three-point stencil for the problem (5)–(7) on a non-uniform grid $\omega = \hat\omega_h \times \hat\omega_\tau$, we will take in account the relation [9]

$$u_{(2)\bar x_1 \hat x_1} - \frac{\partial^2 u(\bar x, t)}{\partial x_1^2} = O(\hbar_1^2 + \hbar_2^2), \quad u_{(1)\bar x_2 \hat x_2} - \frac{\partial^2 u(\bar x, t)}{\partial x_2^2} = O(\hbar_1^2 + \hbar_2^2). \tag{8}$$

The following notation is used here:

$$y_{\bar x_\alpha \hat x_\alpha} = \frac{1}{\hbar_\alpha}(y_{x_\alpha} - y_{\bar x_\alpha}), \quad y_{x_\alpha} = \frac{1}{h_{\alpha+}}(y^{(+1_\alpha)} - y), \quad y_{\bar x_\alpha} = \frac{1}{h_\alpha}(y - y^{(-1_\alpha)}),$$

$$y_{(1)} = y(x^2) = y + \delta_1^+ y_{x_1} + \delta_1^- y_{\bar x_1}, \quad y_{(2)} = y(x^1) = y + \delta_2^+ y_{x_2} + \delta_2^- y_{\bar x_2},$$

$$\bar x = (\bar x_1, \bar x_2)^2, \quad \bar x_\alpha = x_\alpha + \tilde h_\alpha, \quad \tilde h_\alpha = (h_{\alpha+} - h_\alpha)/3, \quad \hbar_\alpha = 0.5(h_\alpha + h_{\alpha+}),$$

$$x^1 = (x_1, \bar x_2), \quad x^2 = (\bar x_1, x_2), \quad \bar g = g(\bar x), \quad \delta_\alpha^\pm = 0.5(\tilde h_\alpha \pm |\tilde h_\alpha|), \quad \alpha = 1,2.$$

We note that the values of the required grid function $y$ at the non-node points $x^1$, $x^2$ are averaged according to second-order approximations. Expressions (8) mean that with respect to an uncountable point $\bar x$ (in the case of a uniform grid $\bar x \equiv x$), they approximate the second partial derivative $u_{(3-\alpha)\bar x_\alpha x_\alpha}$ with respect to the component $x_\alpha$, $\alpha = 1,2$ with the second order. To construct monotone difference schemes of the second order of approximation for Eq. (5) on a non-uniform grid $\omega = \hat\omega_h \times \hat\omega_\tau$, we use the identity $(ku')' = ku'' + k'u'$ and the regularization principle. Equation (5) can be written in the non-divergent form:

$$\frac{\partial u}{\partial t} = \sum_{\alpha=1}^{2} k_\alpha(u)\frac{\partial^2 u}{\partial x_\alpha^2} + \sum_{\alpha=1}^{2}\left(\frac{\partial k_\alpha(u)}{\partial x_\alpha} + v_\alpha(u)\right)\frac{\partial u}{\partial x_\alpha} - q(x)u + f(x,t), \quad x \in \Omega, \quad t \in (0,T]. \tag{9}$$

For the numerical solution of the problem (9), (6), (7) we construct the difference scheme:

$$
\begin{gathered}
y_{(\delta)t} = \sum_{\alpha=1}^{2} \overline{\kappa}_\alpha \overline{k}_\alpha \hat{y}_{(3-\alpha)\overline{x}_\alpha \hat{x}_\alpha} + \sum_{\alpha=1}^{2} (\overline{r}_\alpha^{+} \hat{y}_{(3-\alpha)x_\alpha} + \overline{r}_\alpha^{-} \hat{y}_{(3-\alpha)\overline{x}_\alpha}) - \overline{q}\hat{y}_{(\delta_*)} + \hat{\overline{f}}, \quad x \in \hat{\omega}_h, \\
y_{0i_2}^{n+1} = \mu_1(x_2^{i_2}, t_{n+1}), \quad y_{N_1 i_2}^{n+1} = \mu_2(x_2^{i_2}, t_{n+1}), \quad i_2 = \overline{1, N_2 - 1}, \\
y_{i_1 0}^{n+1} = \mu_3(x_1^{i_1}, t_{n+1}), \quad y_{i_1 N_2}^{n+1} = \mu_4(x_1^{i_1}, t_{n+1}), \quad i_1 = \overline{1, N_1 - 1}, \\
y_{i_1 i_2}^{0} = u_0(x_1^{i_1}, x_2^{i_2}), \quad i_\alpha = \overline{0, N_\alpha}, \quad \alpha = 1, 2,
\end{gathered} \tag{10}
$$

where

$$
y_t = \frac{\hat{y} - y}{\tau_+}, \quad \hat{y} = y(x_{i_1 i_2}, t_{n+1}), \quad \tau_+ = \tau_{n+1},
$$

$$
y_{(\delta)} = y + \sum_{\alpha=1}^{2} (\delta_\alpha^{+} y_{x_\alpha} + \delta_\alpha^{-} y_{\overline{x}_\alpha}), \quad y_{(\delta_*)} = y + \sum_{\alpha=1}^{2} (\delta_\alpha^{-} y_{x_\alpha} + \delta_\alpha^{+} y_{\overline{x}_\alpha}),
$$

$$
\overline{r}_\alpha^{+} = 0.5(\overline{r}_\alpha + |\overline{r}_\alpha|) \ge 0, \quad \overline{r}_\alpha^{-} = 0.5(\overline{r}_\alpha - |\overline{r}_\alpha|) \le 0,
$$

$$
\overline{k}_\alpha = k_\alpha(y_{(\delta)}), \quad \overline{r}_\alpha = (k_\alpha(y_{(3-\alpha)}))_{\overset{\Delta}{x}_\alpha} + v_\alpha(y_{(\delta)}),
$$

$$
(k_\alpha(y_{(3-\alpha)}))_{\overset{\Delta}{x}_\alpha} = \frac{(k_\alpha(y_{(3-\alpha)}))_{x_\alpha} + (k_\alpha(y_{(3-\alpha)}))_{\overline{x}_\alpha} + (k_\alpha(y_{(3-\alpha)}))_{\overset{\circ}{x}_\alpha}}{3},
$$

$$
(k_\alpha(y_{(3-\alpha)}))_{x_\alpha} = \frac{k_\alpha(y_{(3-\alpha)}^{+1_\alpha}) - k_\alpha(y_{(3-\alpha)})}{h_+},
$$

$$
(k_\alpha(y_{(3-\alpha)}))_{\overline{x}_\alpha} = \frac{k_\alpha(y_{(3-\alpha)}) - k_\alpha(y_{(3-\alpha)}^{-1_\alpha})}{h},
$$

$$
(k_\alpha(y_{(3-\alpha)}))_{\overset{\circ}{x}_\alpha} = \frac{k_\alpha(y_{(3-\alpha)}^{+1_\alpha}) - k_\alpha(y_{(3-\alpha)}^{-1_\alpha})}{h + h_+},
$$

$$
\overline{\kappa}_\alpha = \frac{1}{1 + \overline{R}_\alpha}, \quad \overline{R}_\alpha = \overline{b}_\alpha^{+} \frac{h_{\alpha+} + 2h_\alpha}{6} - \overline{b}_\alpha^{-} \frac{2h_{\alpha+} + h_\alpha}{6}, \quad \overline{b}_\alpha^{\pm} = \frac{\overline{r}_\alpha^{\pm}}{\overline{k}_\alpha}.
$$

## 4. THE APPROXIMATION ERROR

Write the approximation error of the difference scheme (10)

$$
\Psi(\overline{x}, \hat{t}) = -u_{(\delta)t} + \sum_{\alpha=1}^{2} \overline{\kappa}_\alpha \overline{k}_\alpha \hat{u}_{(3-\alpha)\overline{x}_\alpha \hat{x}_\alpha} + \sum_{\alpha=1}^{2} (\overline{r}_\alpha^{+} \hat{u}_{(3-\alpha)x_\alpha} + \overline{r}_\alpha^{-} \hat{u}_{(3-\alpha)\overline{x}_\alpha}) - \overline{q}\hat{u}_{(\delta_*)} + \hat{\overline{f}}
$$

in the form

$$
\begin{gathered}
\Psi = \sum_{\alpha=1}^{2} \psi_\alpha^{D} + \sum_{\alpha=1}^{2} \psi_\alpha^{C} + \sum_{\alpha=1}^{2} (\overline{\kappa}_\alpha - 1)\overline{k}_\alpha \frac{\partial^2 u(\overline{x}, \hat{t})}{\partial x_\alpha^2} \\
+ \sum_{\alpha=1}^{2} (\overline{k}_\alpha - k_\alpha(u(\overline{x}, t))) \frac{\partial^2 u(\overline{x}, \hat{t})}{\partial x_\alpha^2} - \overline{q}(\hat{u}_{(\delta_*)} - \hat{u}(\overline{x})) - \left( u_{(\delta)t} - \frac{\partial u(\overline{x}, \hat{t})}{\partial t} \right),
\end{gathered}
$$

where

$$
\psi_\alpha^{D} = \overline{k}_\alpha \left( \hat{u}_{(3-\alpha)\overline{x}_\alpha \hat{x}_\alpha} - \frac{\partial^2 u(\overline{x}, \hat{t})}{\partial x_\alpha^2} \right), \quad \alpha = 1, 2,
$$

$$
\psi_\alpha^{C} = \overline{r}_\alpha^{+} (\hat{u}_{(3-\alpha)} - \hat{u}(x^\alpha))_{x_\alpha} + \overline{r}_\alpha^{-} (\hat{u}_{(3-\alpha)} - \hat{u}(x^\alpha))_{\overline{x}_\alpha}
$$

$$+\bar{r}_\alpha^+(\hat{u}(x^\alpha))_{x_\alpha}+\bar{r}_\alpha^-(\hat{u}(x^\alpha))_{\bar{x}_\alpha}-\left(\frac{\partial k_\alpha(u(\bar{x},t))}{\partial x_\alpha}+v_\alpha(u(\bar{x},t))\right)\frac{\partial u(\bar{x},\hat{t})}{\partial x_\alpha}.$$

On the basis of relations

$$u_t(\bar{x},t)-u_{(\delta)t}=O(\hbar_1^2+\hbar_2^2),\quad \frac{\partial u(\bar{x},\hat{t})}{\partial t}-u_t(\bar{x},t)=O(\tau_+)$$

we get that

$$\frac{\partial u(\bar{x},\hat{t})}{\partial t}-u_{(\delta)t}=O(\hbar_1^2+\hbar_2^2+\tau_+). \tag{11}$$

It follows from (8) that

$$\psi_\alpha^D=O(\hbar_1^2+\hbar_2^2),\quad \alpha=1,2. \tag{12}$$

It is easy to show (see, e.g., [9]) that

$$\bar{k}_\alpha=k_\alpha(u(\bar{x},t))+O(\hbar_1^2+\hbar_2^2), \tag{13}$$

$$\bar{r}_\alpha=\frac{\partial k_\alpha(u(\bar{x},t))}{\partial x_\alpha}+v_\alpha(u(\bar{x},t))+O(\hbar_1^2+\hbar_2^2). \tag{14}$$

Now we prove that

$$\psi_\alpha^C=\bar{R}_\alpha\bar{k}_\alpha\frac{\partial^2 u(\bar{x},\hat{t})}{\partial x_\alpha^2}+O(\hbar_1^2+\hbar_2^2). \tag{15}$$

To do this, we use the following expansions:

$$u_{x_{3-\alpha}}=\frac{\partial u}{\partial x_{3-\alpha}}+\frac{h_{3-\alpha+}}{2}\overline{\frac{\partial^2 u}{\partial x_{3-\alpha}^2}},\quad u_{\bar{x}_{3-\alpha}}=\frac{\partial u}{\partial x_{3-\alpha}}-\frac{h_{3-\alpha}}{2}\overline{\frac{\partial^2 u}{\partial x_{3-\alpha}^2}},$$

where the bar above means that the values of the arguments at the corresponding intermediate points (in this case, on the intervals $(x_{3-\alpha},x_{3-\alpha}+h_{3-\alpha+})$ and $(x_{3-\alpha}-h_{3-\alpha},x_{3-\alpha})$) are taken. In view of this,

$$u_{(3-\alpha)}=u+\tilde{h}_{3-\alpha}\frac{\partial u}{\partial x_{3-\alpha}}+\eta_0(x),$$

$$\eta_0(x)=\frac{\delta_{3-\alpha}^+h_{3-\alpha+}}{2}\overline{\frac{\partial^2 u}{\partial x_{3-\alpha}^2}}-\frac{\delta_{3-\alpha}^-h_{3-\alpha}}{2}\overline{\frac{\partial^2 u}{\partial x_{3-\alpha}^2}}=O(\hbar_{3-\alpha}^2).$$

Given that

$$u(x^\alpha)=u+\tilde{h}_{3-\alpha}\frac{\partial u}{\partial x_{3-\alpha}}+\eta_1(x),\quad \eta_1(x)=\frac{\tilde{h}_{3-\alpha}^2}{2}\overline{\frac{\partial^2 u}{\partial x_{3-\alpha}^2}},$$

for $\xi_1=(u_{(3-\alpha)}-u(x^\alpha))_{x_\alpha}$, $\xi_2=(u_{(3-\alpha)}-u(x^\alpha))_{\bar{x}_\alpha}$ we obtain the representation

$$\xi_1=\eta_{2x_\alpha},\quad \xi_2=\eta_{2\bar{x}_\alpha},\quad \eta_2(x)=\eta_0(x)-\eta_1(x)=O(\hbar_{3-\alpha}^2).$$

As

$$|\eta_{2x_\alpha}|=\left|\frac{1}{h_{\alpha+}}\int\limits_{x_\alpha}^{x_\alpha+h_{\alpha+}}\frac{\partial\eta_2(\zeta,x_{3-\alpha})}{\partial\zeta}d\zeta\right|\le\hbar_{3-\alpha}^2\left\|\frac{\partial^3 u}{\partial x_\alpha\partial x_{3-\alpha}^2}\right\|_{C(\bar{\Omega})},$$

$$|\eta_{2\bar{x}_\alpha}|=\left|\frac{1}{h_\alpha}\int\limits_{x_\alpha-h_\alpha}^{x_\alpha}\frac{\partial\eta_2(\zeta,x_{3-\alpha})}{\partial\zeta}d\zeta\right|\le\hbar_{3-\alpha}^2\left\|\frac{\partial^3 u}{\partial x_\alpha\partial x_{3-\alpha}^2}\right\|_{C(\bar{\Omega})},$$

then

$$\xi_1=O(\hbar_{3-\alpha}^2),\quad \xi_2=O(\hbar_{3-\alpha}^2). \tag{16}$$

Further, using the expansion of the function in a Taylor series

$$(u(x^\alpha))_{x_\alpha} = \frac{\partial u(\bar{x})}{\partial x_\alpha} + \frac{h_{\alpha+} + 2h_\alpha}{6}\frac{\partial^2 u(\bar{x})}{\partial x_\alpha^2} + O(\hbar_\alpha^2),$$

$$(u(x^\alpha))_{\bar{x}_\alpha} = \frac{\partial u(\bar{x})}{\partial x_\alpha} - \frac{2h_{\alpha+} + h_\alpha}{6}\frac{\partial^2 u(\bar{x})}{\partial x_\alpha^2} + O(\hbar_\alpha^2),$$

we conclude that

$$\bar{r}_\alpha^+(u(x^\alpha))_{x_\alpha} + \bar{r}_\alpha^-(u(x^\alpha))_{\bar{x}_\alpha} = \bar{r}_\alpha\frac{\partial u(\bar{x})}{\partial x_\alpha} + \bar{R}_\alpha\bar{k}_\alpha\frac{\partial^2 u(\bar{x})}{\partial x_\alpha^2} + O(\hbar_\alpha^2). \tag{17}$$

From (14), (16), (17) we obtain (15). By virtue of (11)–(13), (15) we get

$$\Psi = \sum_{\alpha=1}^{2}\frac{\bar{R}_\alpha^2}{1+\bar{R}_\alpha}\bar{k}_\alpha\frac{\partial^2 u(\bar{x})}{\partial x_\alpha^2} + O(\hbar_1^2 + \hbar_2^2) = O(\hbar_1^2 + \hbar_2^2).$$

Let $|h| = (h_1^2 + h_2^2)^{1/2}$, $h_\alpha = \max\limits_{i_\alpha} h_\alpha^{i_\alpha}$, $\alpha = 1,2$. Thus we following theorem is valid.

**Theorem 1.** *The approximation error of the finite-difference scheme* (10) *is a second-order small quantity with respect to* $|h|$.

## 5. MONOTONICITY, TWO-SIDED AND A PRIORI ESTIMATES OF APPROXIMATE SOLUTION

For the sake of simplicity of further investigations, we confine ourselves to considering the case of thickening of grids with respect to the variables $x_1$, $x_2$ to the beginning of the corresponding segment. Then, $h_\alpha^{i_\alpha+1} - h_\alpha^{i_\alpha} > 0$, $\delta_\alpha^- = 0$, $\alpha = 1,2$ and Scheme (10) takes a simpler form

$$\begin{aligned} y_t + \frac{h_{1+} - h_1}{3}y_{tx_1} + \frac{h_{2+} - h_2}{3}y_{tx_2} &= \sum_{\alpha=1}^{2}\bar{\kappa}_\alpha\bar{k}_\alpha\left(\hat{y} + \frac{h_{3-\alpha+} - h_{3-\alpha}}{3}\hat{y}_{x_{3-\alpha}}\right)_{\bar{x}_\alpha\hat{x}_\alpha} \\ &+ \sum_{\alpha=1}^{2}\bar{r}_\alpha^+\left(\hat{y} + \frac{h_{3-\alpha+} - h_{3-\alpha}}{3}\hat{y}_{x_{3-\alpha}}\right)_{x_\alpha} + \sum_{\alpha=1}^{2}\bar{r}_\alpha^-\left(\hat{y} + \frac{h_{3-\alpha+} - h_{3-\alpha}}{3}\hat{y}_{x_{3-\alpha}}\right)_{\bar{x}_\alpha} \\ &- q(\bar{x})\left(\hat{y} + \frac{h_{1+} - h_1}{3}\hat{y}_{\bar{x}_1} + \frac{h_{2+} - h_2}{3}\hat{y}_{\bar{x}_2}\right) + f(\bar{x},\hat{t}), \quad x \in \hat{\omega}_h. \end{aligned} \tag{18}$$

To apply the maximum principle, we reduce the scheme (18) to the canonical form (1) and then verify sufficient conditions on the coefficients (2), (3). In the case we are considering, the stencil of the scheme has 8 numbered points and consists of the nodes depicted in Fig. 1. Then for the Scheme (18) we obtain that

$$\sum_{\xi\in\mathcal{M}'(x)} B^n(x,\xi) = \sum_{j=1}^{7} B_j^n.$$

In order to write the coefficients $A^n$, $B^n$, $F^n$, it is necessary to rewrite Scheme (18) in the index form. After elementary transformations we find

$$B_1^n = \tau_+\left(\frac{\bar{\kappa}_2\bar{k}_2}{\hbar_2} - \bar{r}_2^-\right)\frac{h_1 + 2h_{1+}}{3h_2h_{1+}} + \bar{q}\frac{\delta_2^+}{h_2}, \quad B_2^n = \tau_+\left(\frac{\bar{\kappa}_2\bar{k}_2}{\hbar_2} - \bar{r}_2^-\right)\frac{\delta_1^+}{h_2h_{1+}},$$

$$B_3^n = \tau_+\left(\frac{\bar{\kappa}_1\bar{k}_1}{\hbar_1} - \bar{r}_1^-\right)\frac{h_2 + 2h_{2+}}{3h_1h_{2+}} + \bar{q}\frac{\delta_1^+}{h_1}, \quad B_5^n = \tau_+\left(\frac{\bar{\kappa}_1\bar{k}_1}{\hbar_1} - \bar{r}_1^-\right)\frac{\delta_2^+}{h_1h_{2+}},$$

$$B_4^n = -\frac{\delta_1^+}{h_{1+}} + \frac{\tau_+}{h_{1+}h_{2+}}\left(\frac{\bar{\kappa}_1\bar{k}_1(h_2 + 2h_{2+})}{3\hbar_1} + \frac{\bar{r}_1^+(h_2 + 2h_{2+})}{3} - \frac{\delta_1^+}{h_2}(2\bar{\kappa}_2\bar{k}_2 + \bar{r}_2^+h_2 - \bar{r}_2^-h_{2+})\right),$$

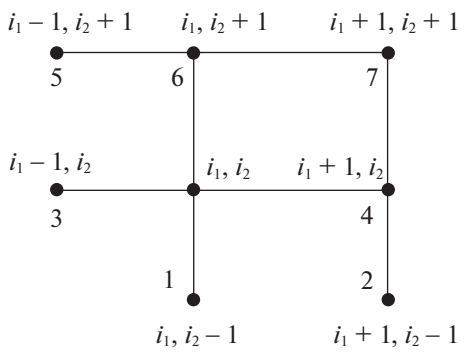


**Fig. 1.** 8-point stencil of the difference Scheme (12).

$$B_6^n = -\frac{\delta_2^+}{h_{2+}} + \frac{\tau_+}{h_{1+}h_{2+}}\left(\frac{\overline{\kappa}_2\overline{k}_2(h_1+2h_{1+})}{3\hbar_2} + \frac{\overline{r}_2^+(h_1+2h_{1+})}{3} - \frac{\delta_2^+}{h_1}(2\overline{\kappa}_1\overline{k}_1 + \overline{r}_1^+h_1 - \overline{r}_1^-h_{1+})\right),$$

$$B_7^n = \frac{\tau_+}{h_{1+}h_{2+}}\left(\frac{\overline{\kappa}_1\overline{k}_1\delta_2^+}{\hbar_1} + \frac{\overline{\kappa}_2\overline{k}_2\delta_1^+}{\hbar_2} + \overline{r}_1^+\delta_2^+ + \overline{r}_2^+\delta_1^+\right), \quad F^n = y_{(\delta)}^n + \tau_+ f(\overline{x},\hat{t}),$$

$$A^n = 1 + \tau_+\overline{q} + \sum_{j=1}^{7} B_j^n, \quad D^n = A^n - \sum_{j=1}^{7} B_j^n = 1 + \tau_+\overline{q}.$$

Now we need to find a condition when $\overline{k}_{\alpha,i_1i_2}^n > 0$ for every $i_\alpha = \overline{1, N-1}$, $\alpha = 1,2$, $n = \overline{0, N_0}$. It is obvious that for $n = 0$ we have $\overline{k}_{\alpha,i_1i_2}^0 = k_\alpha(y_{(\delta)i_1i_2}^0)$, where

$$y_{(\delta)i_1i_2}^0 = u_{0,i_1i_2} + \delta_1^+(u_{0,i_1i_2})_{x_1} + \delta_2^+(u_{0,i_1i_2})_{x_2} = \left(1 - \frac{\tilde{h}_1}{h_{1+}} - \frac{\tilde{h}_2}{h_{2+}}\right)u_{0,i_1i_2} + \frac{\tilde{h}_1}{h_{1+}}u_{0,i_1+1i_2} + \frac{\tilde{h}_2}{h_{2+}}u_{0,i_1i_2+1}.$$

As

$$0 < \frac{\tilde{h}_1}{h_{1+}} < \frac{1}{3}, \quad 0 < \frac{\tilde{h}_2}{h_{2+}} < \frac{1}{3}, \quad 1 - \frac{\tilde{h}_1}{h_{1+}} - \frac{\tilde{h}_2}{h_{2+}} > 0,$$

then $\min\limits_{x\in\Omega} u_0(x) \le y_{(\delta)i_1i_2}^0 \le \max\limits_{x\in\Omega} u_0(x)$, or $y_{(\delta)i_1i_2}^0 \in [m_1, m_2]$, i.e. $\overline{k}_{\alpha,i_1i_2}^0 = k_\alpha(y_{(\delta)i_1i_2}^0) > 0$. Suppose that for an arbitrary $n$, the inequality $\overline{k}_{\alpha,i_1i_2}^n > 0$ is also true. Then from this assumption we have $B_j^n > 0$, $j = 1,2,3,5,7$. The remaining coefficients $B_j^n > 0$, $j = 4,6$ are positive if the following conditions are satisfied

$$\frac{h_{1+} - h_1}{\hbar_1} < \frac{3\overline{\kappa}_1^n\overline{k}_1^n}{2\overline{\kappa}_2^n\overline{k}_2^n + \overline{r}_2^{n+}h_2 - \overline{r}_2^{n-}h_{2+}}\left(\frac{\hbar_2}{\hbar_1}\right)^2, \tag{19}$$

$$\frac{h_{2+} - h_2}{\hbar_2} < \frac{3\overline{\kappa}_2^n\overline{k}_2^n}{2\overline{\kappa}_1^n\overline{k}_1^n + \overline{r}_1^{n+}h_1 - \overline{r}_1^{n-}h_{1+}}\left(\frac{\hbar_1}{\hbar_2}\right)^2, \tag{20}$$

$$\tau_+ \ge \max\{\|\Delta t_1^n\|_C, \|\Delta t_2^n\|_C\}, \tag{21}$$

where

$$\Delta t_1^n = \frac{h_{2+}(h_{1+} - h_1)}{3}\left(\frac{\overline{\kappa}_1^n\overline{k}_1^n(h_2 + 2h_{2+})}{3\hbar_1} - \frac{\delta_1^+}{h_2}(2\overline{\kappa}_2^n\overline{k}_2^n + \overline{r}_2^{n+}h_2 - \overline{r}_2^{n-}h_{2+})\right)^{-1},$$

$$\Delta t_2^n = \frac{h_{1+}(h_{2+} - h_2)}{3}\left(\frac{\overline{\kappa}_2^n\overline{k}_2^n(h_1 + 2h_{1+})}{3\hbar_2} - \frac{\delta_2^+}{h_1}(2\overline{\kappa}_1^n\overline{k}_1^n + \overline{r}_1^{n+}h_1 - \overline{r}_1^{n-}h_{1+})\right)^{-1}.$$

On the basis of the estimate (4) of Lemma for an arbitrary $t = t_n \in \omega_\tau$ and all $i_\alpha = \overline{0, N_\alpha}$, $\alpha = 1,2$ we have

$$\min\left\{\mu_1^{n+1},\mu_2^{n+1},\mu_3^{n+1},\mu_4^{n+1}, \min_{\substack{1\le i_1\le N_1-1\\1\le i_2\le N_2-1}}\left(\frac{y^n_{(\delta)i_1i_2}+\tau_{n+1}\overline{f}^{n+1}_{i_1i_2}}{1+\tau_{n+1}\overline{q}_{i_1i_2}}\right)\right\} \le y^{n+1}_{i_1i_2}$$
$$\le \max\left\{\mu_1^{n+1},\mu_2^{n+1},\mu_3^{n+1},\mu_4^{n+1}, \min_{\substack{1\le i_1\le N_1-1\\1\le i_2\le N_2-1}}\left(\frac{y^n_{(\delta)i_1i_2}+\tau_{n+1}\overline{f}^{n+1}_{i_1i_2}}{1+\tau_{n+1}\overline{q}_{i_1i_2}}\right)\right\}. \tag{22}$$

From obvious inequalities $\min\limits_{\substack{1\le i_1\le N_1-1\\1\le i_2\le N_2-1}}(y^n_{(\delta)i_1i_2}) \ge \min\limits_{\substack{1\le i_1\le N_1-1\\1\le i_2\le N_2-1}} y^n_{i_1i_2}$, $\max\limits_{\substack{1\le i_1\le N_1-1\\1\le i_2\le N_2-1}}(y^n_{(\delta)i_1i_2}) \le \max\limits_{\substack{1\le i_1\le N_1-1\\1\le i_2\le N_2-1}} y^n_{i_1i_2}$ from (22) it follows that

$$\min\left\{0,\mu_1^{n+1},\mu_2^{n+1},\mu_3^{n+1},\mu_4^{n+1}, \min_{\substack{1\le i_1\le N_1-1\\1\le i_2\le N_2-1}} y^n_{i_1i_2}+\tau_{n+1}\inf_{(x,t)\in Q_T} f(x,t)\right\} \le y^{n+1}_{i_1i_2}$$
$$\le \max\left\{0,\mu_1^{n+1},\mu_2^{n+1},\mu_3^{n+1},\mu_4^{n+1}, \min_{\substack{1\le i_1\le N_1-1\\1\le i_2\le N_2-1}} y^n_{i_1i_2}+\tau_{n+1}\sup_{(x,t)\in Q_T} f(x,t)\right\}. \tag{23}$$

By applying induction on $n$, the required two-sided estimates in terms of the input data are derived from (23) without assuming the sign-definiteness of input data

$$m_1^{n+1} \le y^{n+1}_{i_1i_2} \le m_2^{n+1}, \quad i_\alpha = 0,1,\dots N_\alpha, \quad \alpha = 1,2, \tag{24}$$

where

$$m_1^{n+1} = \min\left\{0,\mu_{\min}(x,t), \min_{x\in\overline{\Omega}} u_0(x)\right\} + t_{n+1}\min\left\{0, \inf_{(x,t)\in Q_T} f(x,t)\right\} \ge m_1,$$
$$m_2^{n+1} = \max\left\{0,\mu_{\max}(x,t), \max_{x\in\overline{\Omega}} u_0(x)\right\} + t_{n+1}\max\left\{0, \sup_{(x,t)\in Q_T} f(x,t)\right\} \le m_2.$$

From (24) we obtain that $y^{n+1}_{i_1i_2} \in [m_1, m_2]$, $i_\alpha = 0,1,\dots N_\alpha$, $\alpha = 1,2$, i.e. $\overline{k}^{n+1}_{\alpha,i_1i_2} = k_\alpha(y^{n+1}_{(\delta)i_1i_2}) > 0$. Thus, inequalities (19)–(21) guarantee the fulfillment of the monotonicity condition (2)–(3). Then the following statement holds.

**Theorem 2.** *Suppose that conditions* (19)–(21) *are fulfilled. Then difference scheme* (18) *is monotone and its solution satisfies the two-sided estimates of the form* (24).

**Theorem 3.** *Suppose that conditions* (19)–(21) *are satisfied. Then for the solution of the difference scheme* (18) *for any* $t_n \in \hat{\omega}_\tau$ *the following a priori estimate is valid*:

$$\|y(t_{n+1})\|_{\overline{C}} \le \max\left\{\|u_0\|_{\overline{C}}, \max_{1\le k\le n+1}\|\mu(t_k)\|_{C_\gamma}\right\} + t_{n+1}\max_{1\le k\le n+1}\|\overline{f}(t_k)\|_C, \tag{25}$$

*where*

$$\|v\|_{\overline{C}} = \max_{x\in\overline{\omega}_h}|v(x)|, \quad \|v\|_{C_\gamma} = \max_{x\in\gamma_h}|v(x)|,$$
$$\|\mu(t_k)\|_{C_\gamma} = \max\{\|\mu_1(t_k)\|_{C_\gamma}, \|\mu_2(t_k)\|_{C_\gamma}, \|\mu_3(t_k)\|_{C_\gamma}, \|\mu_4(t_k)\|_{C_\gamma}\}.$$

**Proof.** On the basis of Corollary, we conclude that

$$\|y(t_{n+1})\|_{\overline{C}} = \max\left\{\|\mu(t_{n+1})\|_{C_\gamma}, \left\|\frac{F^n}{D^n}\right\|_C\right\}.$$

Substituting the estimate

$$\|F^n\|_C = \left\|\frac{y^n_{(\delta)}+\tau_{n+1}\overline{f}(t_{n+1})}{1+\tau_{n+1}\overline{q}}\right\|_C \le \|y^n_{(\delta)}\|_C + \tau_{n+1}\|\overline{f}(t_{n+1})\|_C \le \|y(t_n)\|_{\overline{C}} + \tau_{n+1}\|\overline{f}(t_{n+1})\|_C$$

in the last inequality, we find the chain of relations

$$\|y(t_{n+1})\|_{\bar{C}} \le \max\{\|\mu(t_{n+1})\|_{C_\gamma}, \|y(t_n)\|_{\bar{C}} + \tau_{n+1}\|\bar{f}(t_{n+1})\|_C\}$$

$$\le \max\{\|\mu(t_{n+1})\|_{C_\gamma}, \|\mu(t_n)\|_{C_\gamma} + \tau_{n+1}\|\bar{f}(t_{n+1})\|_C, \|y(t_{n-1})\|_{\bar{C}} + (\tau_n\|\bar{f}(t_n)\|_C + \tau_{n+1}\|\bar{f}(t_{n+1})\|_C)\}$$

$$\le \ldots \le \max\left\{\max_{1\le k\le n+1}\|\mu(t_k)\|_{C_\gamma} + \sum_{k=0}^{n}\tau_{k+1}\|\bar{f}(t_{k+1})\|_C, \|u_0\|_{\bar{C}} + \sum_{k=0}^{n}\tau_{k+1}\|\bar{f}(t_{k+1})\|_C\right\}.$$

Given that

$$\sum_{k=0}^{n}\tau_{k+1}\|\bar{f}(t_{k+1})\|_C \le \sum_{k=0}^{n}\tau_{k+1}\max_{1\le k\le n+1}\|f(t_k)\|_C = t_{n+1}\max_{1\le k\le n+1}\|\bar{f}(t_k)\|_C,$$

we obtain the required relation (25).

## 6. CONCLUSIONS

In this paper, we proposed a new second-order in space monotone difference scheme on non-uniform grid that approximates the Dirichlet problem for 2D quasi-linear convection-diffusion equation with unbounded nonlinearity. Under several constraints on the grid, two-side estimates of the solution of the scheme are established. Such estimates permit not only to prove the non-negativity of the exact solution, important in physical problems, but also to find sufficient conditions on the input data when the nonlinear problem is parabolic. As a result, a priori estimates of the approximate solution in the grid norm $C$ that depend on the initial and boundary conditions and on the right-hand side only are proved. The proposed schemes are not absolutely stable, and a detailed stability analysis, involving relations of integration steps, is an important issue which we plan to study in the future along with convergence analysis [10–12] and applications.

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